

Reproducibility Information

Environment

Python:

Version: Python 3.8 or later (tested with 3.10 and 3.11)

Dependencies: None (standard library only for core functionality)

Optional: streamlit, pandas (for web UI only)

LibreLane / OpenROAD:

LibreLane version: v3.0.3 (used for clock_8, clock_10, clock_11, clock_12 via Docker)

OpenROAD version: Bundled within the LibreLane v3.0.3 Docker image

Note for clock_25/clock_30: These reports predate the Docker-based executions. The exact

LibreLane/OpenROAD version used to generate them was not recorded at the time. The PDK

and standard cells are the same (sky130_fd_sc_hd).

LibreLane/OpenROAD is only required to generate new reports. The parser and assistant work on pre-existing report artifacts without it.

PDK and Standard Cells:

PDK: sky130

Standard cell library: sky130_fd_sc_hd (confirmed from report content)

Exact PDK commit: not recorded

Operating System:

Reports were generated on a Linux-based system (Docker container).

The assistant and parser are cross-platform (Windows, Linux, macOS).

Docker:

Image: ghcr.io/librelane/librelane:3.0.3

The run_experiments.py script supports a --dockerized flag for container-based runs.

How to Reproduce

1. Parse existing reports (no LibreLane needed):

```
python scripts/parse_existing_runs.py
```

This reads all report files mapped via results/important_reports/run_mapping.json (including metrics_clock11.json, metrics_clock12.json, metrics_run1.json, and metrics_run2.json) and produces:

- results/qor_metrics.csv
- results/parsed_summary.json

2. Run the assistant:

```
python scripts/assistant.py summary
python scripts/assistant.py "fastest setup-clean run"
```

3. Run unit tests:

```
python -m unittest discover -s tests -v
```

4. Run evaluation scenarios:

```
python evaluation/evaluate.py
```

5. Run full project validation:

```
python scripts/validate_project.py
```

6. Generate experiment configs (dry-run):

```
python scripts/run_experiments.py --dry-run
```

7. Run auto-tuning (dry-run):

```
python scripts/auto_tune.py
```

8. Optional - Enable LLM mode:

Set the environment variable OPENAI_API_KEY to your key, then run:

```
python scripts/assistant.py --llm
```

Or set USE_LLM=true along with the API key.

Experiment Coverage

Ten clock-period configurations were explored for PM32. The repository contains:

- 4 completed physical runs (clock_11, clock_12, clock_25, clock_30) with full report artifacts
- 2 execution failures (clock_8, clock_10) where LibreLane was genuinely executed but terminated before final metrics
- 4 config-only (clock_15, clock_18, clock_20, clock_22) where configurations exist but no execution was attempted

LibreLane Execution Details:

Tool: LibreLane v3.0.3 via Docker (ghcr.io/librelane/librelane:3.0.3)

Command format:

```
docker run --rm -v "<project_dir>:/work" -w /work ghcr.io/librelane/librelane:3.0.3 librelane
designs/pm32/config_clock<N>.json
```

PDK: sky130 (sky130_fd_sc_hd standard cells)

Reproducing clock_11 and clock_12:

Requires Docker with the ghcr.io/librelane/librelane:3.0.3 image pulled.

```
docker run --rm -v "$(pwd):/work" -w /work ghcr.io/librelane/librelane:3.0.3 librelane
designs/pm32/config_clock11.json
```

```
docker run --rm -v "$(pwd):/work" -w /work ghcr.io/librelane/librelane:3.0.3 librelane
designs/pm32/config_clock12.json
```

After execution, copy designs/pm32/runs/<RUN_TAG>/final/metrics.json to results/important_reports/ and update run_mapping.json.

The four completed runs are sufficient to reproduce and demonstrate the main software contribution: deterministic report parsing, correct WNS/WS extraction, run comparison, rule-based diagnosis, conversational answers with evidence, and explicit missing-data handling.

The available evidence is not sufficient to claim that the full ten-point clock sweep was completed or that a global optimum was found. The fastest setup-clean run is clock_11 (CLOCK_PERIOD=11 ns, setup slack=1.237 ns), but it has 247 slew violations and very high utilization.

Report Locations

results/important_reports/metrics_clock11.json	OpenROAD metrics for clock_11
results/important_reports/metrics_clock12.json	OpenROAD metrics for clock_12
results/important_reports/metrics_run1.json	OpenROAD metrics for clock_25
results/important_reports/metrics_run2.json	OpenROAD metrics for clock_30
results/important_reports/summary_clock11.rpt	Timing table for clock_11
results/important_reports/summary_clock12.rpt	Timing table for clock_12
results/important_reports/summary_run1.rpt	Timing table for clock_25
results/important_reports/summary_run2.rpt	Timing table for clock_30
results/important_reports/run_mapping.json	File-to-run-ID mapping

Known Missing Artifacts

- Reports for clock_15, clock_18, clock_20, clock_22 are not available (config-only, never executed).
- clock_8 and clock_10 were executed but the flow terminated before producing final metrics.
- Explicit global-routing congestion overflow metrics are not reported by this flow configuration.

Generated Files

All generated files can be reproduced by running the scripts above:

results/qor_metrics.csv	from parse_existing_runs.py
results/parsed_summary.json	from parse_existing_runs.py
results/qor_runs.csv	from run_experiments.py --dry-run
evaluation/results.md	from evaluation/evaluate.py

No Secrets in Repository

No API keys are stored in the repository. LLM functionality requires setting environment variables at runtime. The .gitignore excludes .env and .env.* files.